



 **Time:** 5 minutes

 **Outcome:** Apply the quantity equation, velocity, money growth, price levels, and inflation.

Difficulty:  Intermediate

Quantity Theory of Money



THE CORE IDEA

The quantity equation $M \times V = P \times Y$ links the money stock (M), velocity (V), price level (P), and real output (Y). Velocity is the average number of times money finances final spending in a period. With velocity stable and output determined by productive capacity, sustained money growth above real-output growth raises long-run inflation.



HOW TO RECOGNIZE IT

- Use $M \times V = P \times Y$ to solve for a missing variable.
- With velocity and real output fixed, a proportional money increase produces the same proportional price increase.
- Distinguish the price level P from the value of money $1/P$.

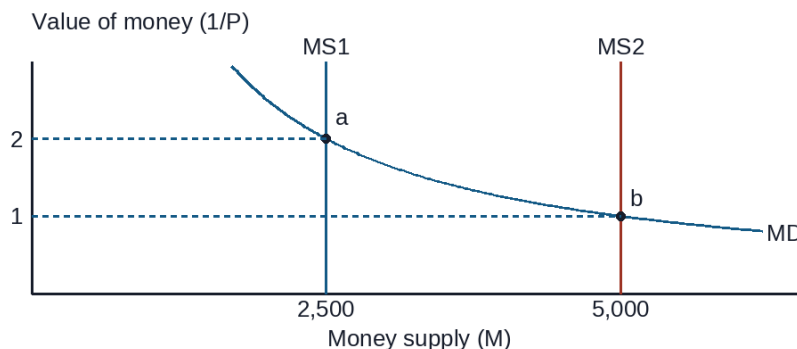


WATCH OUT

Do not assume money growth permanently raises real output. The quantity theory connects persistent money growth to inflation, especially in the long run.



WORKED EXAMPLE: MONEY GROWTH AND INFLATION



Hold $V = 2$ and $Y = 10,000$ fixed. At $M = 2,500$, $P = (2,500 \times 2) / 10,000 = 0.50$. Doubling M to 5,000 raises P to 1.00. The graph plots the reciprocal, $1/P$: the value of money falls from 2 at a to 1 at b. Both calculations describe the same 100% price-level increase.



CHECK YOURSELF

With velocity and real output fixed, money grows by 10%. What happens to the price level in this quantity-theory example, and why does real output not rise?

**READY?**

Return to the game and master this concept.